School of Electronic Engineering and Computer Science
Queen Mary University of London

MSc PROJECT DEFINITION
2025-26

This project definition must be undertaken in consultation with your supervisor. The feasibility of the project should have been assessed and the project aims should be clearly defined.
Submission of this document implies that you have discussed the specification with your supervisor.

Project Title: Auto-Echo: A Framework for Automated Discovery of Memory Hierarchy Latency Patterns

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PROJECT AIMS:
The main aim of this project is to build a Python framework that not only collects echolocation data but also automatically discovers the number and boundaries of memory hierarchy levels without prior knowledge.

PROJECT OBJECTIVES:
- *Automated Memory Hierarchy Discovery -* Develop a system that can automatically determine how many memory levels exist on a machine and identify their boundaries (e.g., L1, L2, L3, DRAM), without manual input.
- *Latency Data Collection and Cleaning -* Implement or adapt an echolocation-style memory probe to collect a large dataset of memory access timings. Clean this data using statistical techniques such as outlier removal and smoothing to ensure it is reliable for analysis.
- *Applying Unsupervised Machine Learning -* Use clustering algorithms (K-Means, Gaussian Mixture Models, DBSCAN) to group latency data so that different memory levels naturally appear as separate clusters, without needing labelled data.

- *Automatic Model Selection* - Introduce methods such as the Elbow Method or Silhouette Score to automatically determine the most appropriate number of clusters, removing the need for manual tuning.
- *Validation Against Known Hardware* - Test the framework on machines with known memory specifications and compare the results to verify accuracy.
- *Reporting Results* - Generate a clear, structured report showing the detected memory levels and their corresponding latency ranges.

Alignment with Programme Learning Outcomes

This project aligns comprehensively with the learning outcomes of the MSc Advanced Computer Science (Software and Data Engineering) programme at QMUL. It engages with core computer science theory through the application of unsupervised machine learning algorithms, drawing directly from the Machine Learning module. The dedicated validation stage, in which the framework's outputs are benchmarked against known hardware specifications, fulfills the testing and evaluation outcome, supported by analytical skills from Data Analytics and Bayesian Decision and Risk Analysis.

METHODOLOGY:

1. The first phase involves a thorough review of the existing literature, most notably the ECOOP 2025 paper "Shouting at Memory: Where Did My Write Go?" to understand the current state of memory echolocation techniques and identify the gap that automation can address.
2. An echolocation-style memory probe will be implemented or adapted in Python to collect a large dataset of memory access latencies from one or more target machines.
3. The raw timing data will then be cleaned using statistical noise filtering techniques, such as outlier removal and moving averages, to ensure the quality and reliability of inputs fed into the subsequent analysis stage.
4. Automatically identify distinct memory hierarchy levels as separable clusters. Model selection techniques, including the Elbow Method and Silhouette Score, will be systematically applied and compared to determine the most accurate and robust method for automatically inferring the number of memory levels.

PROJECT MILESTONES

1. *Literature Review Report:* A review of relevant research papers specifically "Shouting at Memory: Where Did My Write Go?"
2. *Memory Latency Data Collection Module:* A working Python module that probes a target machine and collects a sizable, reproducible dataset of memory access latencies across all hierarchy levels.

3. *Data Preprocessing Pipeline:* A cleaning pipeline that removes outliers and smooths noise from the raw latency data, producing a reliable dataset ready for machine learning analysis.
4. *Clustering and Model Selection Engine:* A clustering module running K-Means, Gaussian Mixture Models, and DBSCAN, with built-in model selection logic that automatically determines the number of memory levels without any manual input.
5. *Validation and Evaluation Report:* A technical report benchmarking the framework's findings against real hardware specifications, identifying which algorithm combination performs most accurately and consistently.
6. *Final Framework and Dissertation:* The complete, documented Auto-Echo pipeline alongside the MSc dissertation, covering the full research narrative from motivation and methodology through to results and conclusions.

REQUIRED KNOWLEDGE/ SKILLS/TOOLS/RESOURCES:

A solid understanding of computer architecture, particularly memory hierarchy design, cache organisation, and the latency characteristics of L1, L2, L3 caches and DRAM is fundamental for this project. In addition, a strong grounding in unsupervised machine learning theory, including clustering algorithms, model selection techniques, knowledge of statistical data preprocessing methods, such as outlier detection and smoothing techniques, is essential for the analytical core of the framework. Finally, familiarity with academic research methodology including critical literature review, experimental design, and structured evaluation is necessary to situate the project within the existing body of work and produce a rigorous dissertation.

Software Tools

- **Python 3.x -** primary programming language for the entire framework including libraries like NumPy, Pandas, Seaborn, Matplotlib, Scikit-learn.
- **Git / GitHub -** for project management and version control.
- **Zenodo Artifact** (associated with the ECOOP 2025 paper) as a reference implementation of the memory echolocation probe

Hardware Resources

Access to one or more physical machines with known, documented memory hierarchy specifications will be required for data collection and validation. At minimum, a machine with a multi-level cache architecture (L1, L2, L3, and DRAM) is needed to generate meaningful latency data and to verify the framework's outputs against ground truth specifications. Access to QMUL laboratory machines or personal hardware meeting these requirements would be sufficient for this project.

Other Resources

Access to academic databases including ACM Digital Library, IEEE Xplore, and Google Scholar will be required for the literature review phase. The ECOOP 2025 paper *"Shouting at Memory: Where Did My Write Go?"* and its associated Zenodo artifact serve as the primary reference and starting point for the project. Supervision and feedback from an assigned QMUL academic supervisor will also constitute an essential resource throughout the research and development process.

TIMEPLAN

- *Phase 1 (Mar 17 – Apr 11)* - Literature review and probe artifact study, to gain a solid research foundation before any coding begins.
- *Phase 2 (Apr 12 – May 12)* - Memory probe implementation and preprocessing pipeline to allow findings from data collection to inform the cleaning strategy.
- *Phase 3 (May 13 – Jun 6)* - Clustering engine and model selection logic, building iteratively on the clean dataset from Phase 2.
- *Phase 4 (Jun 7 – Jun 25)* - Hardware validation and comparative analysis.
- *Phase 5 (May 19 – Jul 9)* - Dissertation writing starts early alongside ML development, with the draft review and submission window in the first week of July.